

Algebraic and Transcendental functions

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Objective: Practice for PrepaTec avanza and Final exam

Algebra skills

Simplify the following expressions,

- 1) $x - [3a + 2(-x + 1)]$,
- 2) $-[3x - 2y + (x + 2y) - 2(x + y) - 3(x + 1)]$,
- 3) $a - (x + y) - 3(x - y) + 2[-(x - 2y) - 2(-x - y)]$.

Find all real zeros of the following polynomials,

- 1) $P(x) = 3x^3 + 5x^2 - 2x - 4$
- 2) $P(x) = 3x^4 - 5x^3 - 16x^2 + 7x + 15$
- 3) $P(x) = x^4 - 6x^3 + 4x^2 + 15x + 4$
- 4) $P(x) = 3x^3 + 4x^2 - x - 2$

Function analysis

Mention the domain and range of the following functions,

- 1) $f(x) = \sqrt{x^2 - 1}$,
- 2) $g(x) = \frac{2}{x(3x + 5)}$,
- 3) $h(x) = -\frac{3}{2}x^2$.

Modelling with math

a) A school fund-raising group sells chocolate bars to help finance a swimming pool for their physical education program. The group finds that when they set their price at x dollars per bar (where $0 < x \leq 5$), their total sales revenue (in dollars) is given by the function $R(x) = -500x^2 + 3000x$.

- Evaluate $R(2)$ and $R(4)$. What do these values represent?
- How many bars they need to sell to get even?

b) A right triangle has one leg twice as long as the other. Find a function that models its perimeter P in terms of the length x of the shorter leg.

c) A Norman window has the shape of a rectangle surmounted by a semi-circle. A Norman window with perimeter 30 ft is to be constructed.

- Find a function that models the area of the window.
- Find the dimensions of the window that admits the greatest amount of light.